

Econ 440: Unconfoundedness-Based Design: Identification

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1 Introduction

Although experimental designs and random assignment are the gold-standard for estimating causal effects, we are not always able to estimate effects in that way. We now move to *non-experimental* designs.

Unconfoundedness-Based Designs are likely the most widely used methods in practice (for good or bad), as they do not require an *Instrument* or a *Quasi-Experiment*, which can be hard to find. However, it is important to understand their assumptions and limitations.

2 Assumptions

All Unconfoundedness-Based Methods are based on the following two Assumptions:

1. *Unconfoundedness*
2. *Overlap*

For the following discussion, let X_i be a set of observed *pre-treatment* variables (or covariates) of unit “ i ” (e.g. i ’s age, gender, race, IQ, income, etc).

2.1 Assumption 1: Unconfoundedness

Assumption 1 (Unconfoundedness). Assume that the treatment T_i is *independent* of the potential outcomes $Y_i(1)$ and $Y_i(0)$ *conditional* on X_i . Formally, assume

$$T_i \perp\!\!\!\perp \{Y_i(1), Y_i(0)\} | X_i$$

Interpretation:

- Treatment Assignment (T_i) is independent of the potential outcomes, **conditional on, or given**, X_i (informally, “after adjusting for X_i ”)
- *Within subpopulations or groups* with the same values of the covariates (X), it is “as if” the treatment (T) had been randomly assigned (RA). That is, *within groups* with the same values of the covariates, it is “as if” we had an experiment: $T_i \perp\!\!\!\perp \{Y_i(1), Y_i(0)\}$.
 - More precisely, T_i is assumed to be determined by unobserved factors **not** correlated with $\{Y_i(1), Y_i(0)\}$.
 - Example: Let Y be earnings and T be “college” $\implies$ “college” is assumed to be determined by factors that affect “college” but do not affect earnings.
- Unconfoundedness assumes that beyond the observed covariates X , there are no other confounders (factors) that affect **both** T and Y .

- We assume that we observe **all** possible confounders (X) $\implies$ there are no unobserved confounders (U).
- In the context of OLS: We assume there is no omitted variable bias.
- Informally, Unconfoundedness says:
 - “We control for every possible confounder”.
 - “After adjusting for covariates (X), it is as if the treatment (T) had been RA.”

Notes on Assumption 1 (Unconfoundedness):

- Assumption 1 is also called: *Selection on Observables*, *Conditional Independence Assumption (CIA)*, or *Exogeneity*.
- Assumption 1 is a *huge assumption*. We claim that we see *all things* that affect **both** T and Y .
- Contrary to the experimental design, in the unconfoundedness design we get a causal effect of T on Y “by assumption”, meaning that we have to assume that we see all confounders.
- Assumption 1 is not testable. We cannot test whether it is true or not.
 - Its validity in any given application must be justified using *economic theory and institutional knowledge*. For instance, we can use economic theory and institutional knowledge to determine if there is a confounder that we are not adjusting for.
 - * For example, based on economic theory, do we have all factors that affect both who attend college (T) and earnings (Y)?
 - Unconfoundedness is *more plausible if we have a large number (or “rich” set) of covariates*. It is more plausible if we control for as many covariates as possible.

Example: Let T be college and let Y be earnings:

- * If we only control for age and gender $\implies$ it is unlikely that Unconfoundedness holds.
- * If we control for age, gender, race, parents' income, parents' education, IQ, zip code where the person grew up, number of siblings, physical and mental health, crime history, etc. $\implies$ It is much more likely that Unconfoundedness holds.
- Key: X must be a set of *pre-treatment* variables. They should not be affected by the treatment variable, otherwise they are *post-treatment* variables, or *intermediate outcomes*.
- In summary: Unconfoundedness states that the treatment (T) is as if it had been Randomly Assigned within groups with the *same* values of X .

$\implies$ We need to compare *similar or comparable* units in terms of X ; that is, units that are alike in terms of the observables X .

Later, we will learn how to adjust for X . First, we discuss how unconfoundedness helps to identify the ATE .

2.2 How Does Unconfoundedness Help to Identify the ATE ?

The key intuition for identification of the ATE under unconfoundedness is that, *within groups with the same values of X* , we “have” (more precisely, assume) T is RA.

- In other words, *within groups of individuals with the same values of X* , we are back to the experimental design!

2.2.1 Aside: Identification of the ATE in the Experimental Design

Before discussing identification of the ATE under unconfoundedness, it is useful to review identification of the ATE in the experimental design.

Recall that the key assumption in the experimental design is that T is Randomly Assigned, which implies that T is independent of the potential outcomes $\{Y(1), Y(0)\}$, or $T \perp\!\!\!\perp \{Y(1), Y(0)\}$.

To mathematically show that the ATE is identified under random assignment of T , write:

$$ATE = E[Y(1) - Y(0)]$$

Using the linearity of expectations, we can write:

$$ATE = E[Y(1)] - E[Y(0)]$$

Next, using the fact that T is independent of the potential outcomes $Y(1)$ and $Y(0)$ (which implies the $T = 1$ and $T = 0$ are “comparable or similar”), we have:

$$ATE = E[Y(1)|T = t] - E[Y(0)|T = t]$$

$$ATE = E[Y(1)|T = 1] - E[Y(0)|T = 0]$$

Finally, using the fact that the outcome we observe, Y , equals $Y(1)$ if $T = 1$ and $Y(0)$ if $T = 0$, we have:

$$ATE = E[Y|T = 1] - E[Y|T = 0]$$

This gives us the ATE as a function of only observed variables (Y, T), which means that the ATE is *identified*, so it can be estimated.

As we know, if T is RA, the ATE equals the expected value of the outcome Y for the treated group ($T = 1$) minus the expected value of the outcome Y for the control group ($T = 0$).

2.2.2 Identification of the ATE in Unconfoundedness-Based Designs

To mathematically show how the ATE can be identified in the unconfoundedness-based design, write:

$$ATE = E[Y(1) - Y(0)]$$

Using the Law of Iterated Expectations we have:

$$ATE = E_X[E[Y(1) - Y(0)|X = x]]$$

Using the linearity of expectations, we can write the inner conditional expectation as:

$$ATE = E_X[E[Y(1)|X = x] - E[Y(0)|X = x]]$$

Next, we use the assumption of unconfoundedness: Within groups with the *same* values of X (i.e., *conditional on X*), T is *independent* of the potential outcomes $Y(1)$ and $Y(0)$ (as in an experiment). Intuitively, within groups with the *same values* of X , the groups

$T = 1$ and $T = 0$ are comparable. Thus, we can write:

$$ATE = E_X[E[Y(1)|T = t, X = x] - E[Y(0)|T = t, X = x]]$$

$$ATE = E_X[E[Y(1)|T = 1, X = x] - E[Y(0)|T = 0, X = x]]$$

Finally, recall that the observed outcome is $Y = \begin{cases} Y(1) & \text{if } T = 1 \\ Y(0) & \text{if } T = 0 \end{cases}$ Therefore:

$$\boxed{ATE = E_X[E[Y|T = 1, X = x] - E[Y|T = 0, X = x]]} \quad (1)$$

This gives us the result that, under unconfoundedness (and as we discuss below, overlap) the ATE is a function of the observed variables $(T, Y, X) \implies$ the ATE is identified, so we can estimate it!

Note: Aside from the use of the Law of Iterated Expectations at the beginning to condition on the covariates X , the steps we followed above are basically the same as those used to identify the ATE in the experimental design in the aside above (section 2.2.1). The main difference is that we now condition on the covariates X , because the main assumption under unconfoundedness is that, after we condition on X , is as if we had an experiment.

Interpretation of the Identification Result in (1): *Within groups with the same values of X (i.e. similar or comparable groups in terms of X), we compare the mean outcome for those with $T = 1$ to the mean outcome for those with $T = 0$ (like in an experiment!). Then, the ATE equals the weighted average of the effects for all the groups with different values of X (this is what the outer expectation $E_X[\cdot]$ in (1) is doing).*

2.2.3 Another Way to Derive the Identification Result in (1)

Another way to think of the identification result in (1), is as if we first estimated the ATE for each of the different groups with the same values of X (within which T is assumed to be RA), and then we averaged the ATE 's of all of these groups.

To see this, let:

$$ATE(x) = E[Y(1) - Y(0)|X = x]$$

be the Average Treatment Effect for the group of individuals with values of the covariates equal to $X = x$.

- For example, if the covariates are *gender* and *white*, we would have four groups: white females, non-white females, white males, and non-white males.
- In this case, $ATE(x)$ would denote the ATE for each of these groups; that is, we would have *four* different $ATE(x)$'s.

Using the linearity of expectations, we have:

$$ATE(x) = E[Y(1)|X = x] - E[Y(0)|X = x]$$

By Unconfoundedness, it is “as if” T was RA conditional on X , or *within* groups with the same values of X (e.g., as if T was RA within each of the four groups in the gender-race example above). Mathematically, unconfoundedness means that T is independent of the potential outcomes $Y(1)$ and $Y(0)$ conditional on X . Thus:

$$ATE(x) = E[Y(1)|T = 1, X = x] - E[Y(0)|T = 0, X = x]$$

Then, since the observed outcome is $Y = \begin{cases} Y(1) & \text{if } T = 1 \\ Y(0) & \text{if } T = 0 \end{cases}$ we have:

$$ATE(x) = E[Y|T = 1, X = x] - E[Y|T = 0, X = x] \tag{2}$$

Hence, the $ATE(x)$ —that is, the effect for the particular group with $X = x$ —is *identified*.

- In the context of the example above, the effect *for each of the four groups* in the example, is identified.

However, keep in mind that we want to estimate the population ATE , rather than the effect for a particular group, the $ATE(x)$. Fortunately, by the Law of Iterated Expectations,

the ATE is just a weighted average of the effects for all the groups ($ATE(x)$). Formally,

$$ATE = E[Y(1) - Y(0)]$$

$$ATE = E_X[\underbrace{E[Y(1) - Y(0)|X = x]}_{=ATE(x)}]$$

$$ATE = E_X[ATE(x)]$$

Thus, combining this result with our identification result for the $ATE(x)$ in (2), we find:

$$ATE = E_X[E[Y|T = 1, X = x] - E[Y|T = 0, X = x]] \quad (3)$$

This is the same result as above.

2.2.4 Example

Assume that the only confounder (observed or unobserved) is Gender:

$$X = \begin{cases} 1 & \text{if Female} \\ 0 & \text{if Male} \end{cases}$$

Therefore, unconfoundedness in this case states that T is Randomly Assigned among females and among males. Then, using the Law of Iterated Expectations we have:

$$ATE = E[Y(1) - Y(0)]$$

$$ATE = E_X[\underbrace{E[Y(1) - Y(0)|X = x]}_{=ATE(x)}] = E_X[ATE(x)]$$

$$ATE = \sum_{x=0}^1 Pr(X = x) \cdot ATE(x)$$

$$ATE = \underbrace{Pr(X = 0)}_{\text{Fraction of Males}} \cdot \underbrace{ATE(0)}_{ATE \text{ for Males}} + \underbrace{Pr(X = 1)}_{\text{Fraction of Females}} \cdot \underbrace{ATE(1)}_{ATE \text{ for Females}}$$

where in the last two lines we expanded the outer expectation $E_X[\cdot]$.

Finally, using our identification result for the $ATE(x)$ in equation (2), we get:

$$\begin{aligned} ATE &= \Pr(X = 0)\{E[Y|T = 1, X = 0] - E[Y|T = 0, X = 0]\} \\ &\quad + \Pr(X = 1)\{E[Y|T = 1, X = 1] - E[Y|T = 0, X = 1]\} \end{aligned}$$

2.3 Assumption 2: Overlap

Assumption 2 (Overlap). For all possible values of the covariates X , there is a positive probability to have individuals with $T = 1$ and $T = 0$. Formally, assume:

$$0 < \Pr(T = 1|X = x) < 1 \text{ for all } x \tag{4}$$

Intuition: For all possible values of the covariates X , there are *both* treated and control units.

More formally: For all possible values of X there is a positive probability to have units with $T=1$ and $T=0$, so that in large (or infinite) samples we have both treated and control units.

2.3.1 Why Do We Need Overlap?

Without Overlap, we cannot estimate the ATE , *even* if Unconfoundedness holds.

Example:

Consider the example in section 2.2.4, where unconfoundedness holds after conditioning on *only one* covariate, gender. Also, assume that all females are always treated (e.g., assume all females attend college, our treatment)

$\implies$ Females always have $T = 1$.

Then:

$$ATE(female) = E[Y|T = 1, X = 1] - \underbrace{E[Y|T = 0, X = 1]}_{\text{no observations!}}$$

$\implies$ We cannot estimate $E[Y|T = 0, X = 1] \implies$ We cannot estimate $ATE(female) \implies$ We cannot estimate the ATE for the population!

Therefore, to identify the ATE we need:

1. Unconfoundedness
2. Overlap

Assumptions 1 (Unconfoundedness) and 2 (Overlap) together are called “Strong Ignorability” (Rosenbaum and Rubin, 1983).

Later in the course we will learn how to check for overlap.

2.4 Identification of the ATE: Conclusion

In summary, as discussed above, we can identify (and thus estimate) the ATE under Assumptions 1 (Unconfoundedness) and 2 (Overlap). In particular, under these two assumptions, we have from equation (1) that the ATE is identified as:

$$ATE = E_X[E[Y|T = 1, X = x] - E[Y|T = 0, X = x]]$$

Given this result, how can we *estimate* the ATE under Assumptions 1 and 2?

- If we only have 1 covariate X and is binary (e.g., gender), then it is easy! See our example in section 2.2.4.
- Challenge: Remember that Unconfoundedness states that we observe **all** possible confounders $\implies$ *Usually we need to control for many covariates (some of them continuous rather than binary)* to argue that unconfoundedness holds $\implies$ It is not that easy to estimate the ATE !
- In the past 20-30 years, there has been an increase in the number of methods/tools available to estimate the ATE under an Unconfoundedness-Based Design.
- Some methods/tools to estimate the ATE under an Unconfoundedness-Based Design:
 - OLS (The classic method, although usually not the best for estimating the ATE . It dates back to Gauss in the 1800's)
 - Matching (1990's and 2000's)
 - Propensity Score (“p-score”) Methods
 - * Matching on the Propensity Score
 - * Weighting (covered in the MSQE)
 - * Others (such as Blocking etc.) (Covered in MSQE)
 - Combining the Above Methods (Covered in MSQE)

3 Identification of the ATT under Unconfoundedness

Now that T is not Randomly Assigned, the overall population is not comparable or similar to those with $T = 1$, and the control group is not comparable to the treatment group.

- In our leading example, when who attends college is not randomly assigned, the group of individuals who attend college is not comparable to the group of individuals who do not attend college, or to the overall population. For example, on average, they have higher ability, socioeconomic status, motivation, better health, etc.

$\implies$ $ATE \neq ATT$ in non-experimental settings.

How do we identify the ATT under Unconfoundedness and Overlap?

We follow steps similar to those used to identify the ATE in Section 2.2.2. The ATT is given by:

$$ATT = E[Y(1) - Y(0)|T = 1]$$

Using the linearity of (conditional) expectations, we have:

$$ATT = \underbrace{E[Y(1)|T = 1]}_{=E[Y|T=1] \Rightarrow \text{Identified!}} - \underbrace{E[Y(0)|T = 1]}_{\text{Missing Counterfactual}} \quad (5)$$

Note: Recall that for the ATT we have only *one* missing counterfactual: we need to learn about the potential outcome under control $Y(0)$ for treated units.

- For the ATE , we also need to learn about the potential outcome under treatment $Y(1)$ for the control units (i.e., $E[Y(1)|T = 0]$). This is *not* needed for the ATT .

Next, we use the Law of Iterated Expectations (LIE) for the second term in (5). Here, we use a slightly different version of the LIE from the one we have used before, as it involves a conditional expectation. We explain this version of the LIE in the following aside.

Aside: Law of Iterated Expectations with Conditional Expectations. Given three random variables W, X and Z , we have:

$$E[W|Z = z] = E_{X|Z=z}[E[W|Z = z, X = x]|Z = z] \quad (6)$$

In words, we can write the conditional mean of W given $Z = z$ (i.e., $E[W|Z = z]$), as the weighted average over all possible values of X **given** $Z = z$ (this is the outer expectation $E_{X|Z=z}[\cdot | Z = z]$) of the conditional means of W given **both** $X = x$ and $Z = z$ (these are the inner expectations $E[W|Z = z, X = x]$).

Intuition: It is *the same* LIE as before, but now within units with $Z = z$.

Example: Think about calculating the GPA of *females* in a class. In this case, it is possible to write the GPA of *females* in a class as:

$$\begin{aligned} \text{GPA of females in a Class} = & (\text{GPA of White females})(\text{Fraction of females who are White}) \\ & + (\text{GPA of Non-White females})(\text{Fraction of females who are Non-White}) \end{aligned}$$

To relate this idea to the LIE formula in (6), let W = GPA of a Student in the class, and let $Z = \begin{cases} 1 & \text{if Female} \\ 0 & \text{if Male} \end{cases}$ and $X = \begin{cases} 1 & \text{if White} \\ 0 & \text{if Non-White} \end{cases}$

Then, we can write the GPA of *females* in a class, or $E[W|Z = 1]$, as:

$$\begin{aligned} E[W|Z = 1] &= E_{X|Z=1}[E[W|Z = 1, X = x]|Z = 1] \\ &= \sum_{x=0}^1 \Pr(X = x|Z = 1) \cdot E[W|Z = 1, X = x] \\ &= \Pr(X = 0|Z = 1)E[W|Z = 1, X = 0] + \Pr(X = 1|Z = 1)E[W|Z = 1, X = 1] \end{aligned}$$

Hence, the LIE with conditional expectations is the same as before, but now *within* units with the same values of $Z = z$ (e.g., females).

End Aside.

We now return to identification of the *ATT* under unconfoundedness and overlap. From equation (5) above, we can write the *ATT* as:

$$ATT = E[Y|T = 1] - E[Y(0)|T = 1]$$

Applying to the second term $E[Y(0)|T = 1]$ (i.e., our missing counterfactual) the LIE for conditional expectations discussed in the Aside above, we obtain:

$$ATT = E[Y|T = 1] - E_{X|T=1}[E[Y(0)|T = 1, X = x]|T = 1]$$

Next, we use the assumption of unconfoundedness to re-write the inner expectation of the second term: Within groups with the *same* values of X (i.e., conditional on X), T is *independent* of the potential outcome $Y(0)$ (as in an experiment). Intuitively, as we have discussed before, within groups with the *same values* of X , the groups $T = 1$ and $T = 0$ are comparable. Thus, we can write:

$$ATT = E[Y|T = 1] - E_{X|T=1}[E[Y(0)|T = 0, X = x]|T = 1]$$

Finally, recall that the outcome we observe, Y , equals $Y(0)$ if $T = 0$, which implies that $E[Y(0)|T = 0, X = x] = E[Y|T = 0, X = x]$. Hence, we obtain:

$$\boxed{ATT = E[Y|T = 1] - E_{X|T=1}[E[Y|T = 0, X = x]|T = 1]} \quad (7)$$

Equation(7) implies that under unconfoundedness and overlap, the ATT is identified (i.e., it can be estimated), as it can be written as a function of observed data (namely, $\{T, Y, X\}$).

Notes:

- The steps we followed above to identify the ATT are basically the same as those we used to identify the ATE in section 2.2.2. The main differences are that **(i)** for the ATT , we focus only on one missing counterfactual (the second term in equation (5)); and **(ii)** all the expectations are conditional on $T = 1$ (since this is the average treatment effect *for the treated*, or ATT).
- Intuition for the term $E_{X|T=1}[E[Y|T = 0, X = x]|T = 1]$ in (7).
 1. Inner expectation $E[Y|T = 0, X = x]$: Within each group with $X = x$ (i.e., within groups with same values of the covariates), we calculate/get the mean of the outcome for the control group. We do this for *all* values of X .
 2. Outer expectation $E_{X|T=1}[\cdot |T = 1]$: Then, we take a weighted average of the conditional means $E[Y|T = 0, X = x]$ using the distribution of X in the treated group (rather than the distribution of X in the whole population).
 - Intuitively, since we want the ATT rather than the ATE , the weighted average in the outer expectation (i.e., $E_{X|T=1}[\cdot |T = 1]$) is over the values of the covariates (X) *in the treated group*, rather than over the values of the covariates *in the whole population* (i.e., $E_X(\cdot)$).
 - Note that this is related to the example we discussed in the Aside above about the Law of Iterated Expectations, where we averaged the GPA of White females and Non-White females over the fraction of *females in the class who are White* and Non-White, **and not** over the fraction of *the whole class* (i.e., including females and males) who are Whites and Non-Whites.

3.1 Example

3.2 Setting

As in section 2.2.4, assume that we have only one confounder, gender, where $X = \begin{cases} 1 & \text{if Female} \\ 0 & \text{if Male} \end{cases}$

Also, suppose that:

- The fraction of females in the *population* is: $\Pr(X = 1) = \frac{1}{2}$. (which implies that the fraction of males is $\Pr(X = 0) = \frac{1}{2}$).
- The fraction of females in the *treated group* is: $\Pr(X = 1|T = 1) = \frac{3}{4}$ (which implies that the fraction of males in the treated group is $\Pr(X = 0|T = 1) = \frac{1}{4}$).
 - To be clear, this is the fraction of treated individuals who are female, *and not* the fraction of females who are treated. (Same comment applies to next bullet.)
- The fraction of females in the *control group* is: $\Pr(X = 1|T = 0) = \frac{1}{5}$ (which implies that the fraction of males in the control group is $\Pr(X = 0|T = 0) = \frac{4}{5}$).

$\implies$ Gender (X) is unbalanced between treatment and control groups!

Finally, suppose that:

- The mean earnings of females in the control group is: $E[Y|T = 0, X = 1] = \$1000$.
- The mean earnings of males in the control group is: $E[Y|T = 0, X = 0] = \$800$.

For simplicity, rather than calculating the ATE and ATT , in this example we focus only on the potential outcome $Y(0)$, specifically, we find $E[Y(0)]$ and $E[Y(0)|T = 1]$, which are needed to calculate the ATE and ATT , respectively.

Note: in this example we are assuming that we *know* the true expectations. Hence, we do not need to estimate $E[Y(0)]$ and $E[Y(0)|T = 1]$; we just need to calculate them.

3.2.1 ATE: Finding $E[Y(0)]$

Recall that the ATE is given by:

$$ATE = E[Y(1)] - E[Y(0)]$$

How do we calculate $E[Y(0)]$?

By the LIE, we have (see also our identification result for the ATE in (1)):

$$E[Y(0)] = E_X[\underbrace{E[Y|T = 0, X = x]}_{\text{Mean of } Y \text{ for those with } T = 0 \text{ and } X = x}] \quad (8)$$

In words: First, we get the mean earnings (Y) of those units with $X = x$ in the control group (in our example, $E[Y|T = 0, X = 1]$ and $E[Y|T = 0, X = 0]$). Second, we average those means over the distribution of X (gender) **in the population**.

In our example (using Equation (8)):

$$\begin{aligned} E[Y(0)] &= E_X[E[Y|T = 0, X = x]] \\ &= \sum_{x=0}^1 \Pr(X = x) \cdot E[Y|T = 0, X = x] \\ &= \underbrace{\Pr(X = 0)}_{=\frac{1}{2}} \cdot \underbrace{E[Y|T = 0, X = 0]}_{=800} + \underbrace{\Pr(X = 1)}_{=\frac{1}{2}} \cdot \underbrace{E[Y|T = 0, X = 1]}_{=1000} \\ E[Y(0)] &= \boxed{\$900} \end{aligned}$$

3.2.2 ATT: Finding $E[Y(0)|T = 1]$

Recall that the ATT is given by:

$$ATT = \underbrace{E[Y(1)|T = 1]}_{=E[Y|T=1]} - E[Y(0)|T = 1]$$

How do we calculate $E[Y(0)|T = 1]$?

By the LIE (see also our identification result in (7)):

$$E[Y(0)|T = 1] = E_{X|T=1}[\underbrace{E[Y|T = 0, X = x]}_{\text{Mean of } Y \text{ for those with } T = 0 \text{ and } X = x} | T = 1] \quad (9)$$

In words: First, we get the mean earnings (Y) of those units with $X = x$ in the control group (in our example, $E[Y|T = 0, X = 1]$ and $E[Y|T = 0, X = 0]$) (note: this is the *same* step as for $E[Y(0)]$ above). Second, we average those means over the distribution of X (gender) **in the treated group**.

Note that for the *ATE* (specifically, for $E[Y(0)]$), we used $E_X(\cdot)$: the distribution of X (gender) in the *population*. For the *ATT* (specifically, for $E[Y(0)|T = 1]$), we used $E_{X|T=1}[\cdot | T = 1]$: the distribution of X in the *treated group*.

In our example (using Equation (9)):

$$\begin{aligned}
E[Y(0)|T = 1] &= E_{X|T=1}[E[Y|T = 0, X = x]|T = 1] \\
&= \sum_{x=0}^1 \text{Pr}(X = x|T = 1) \cdot E[Y|T = 0, X = x] \\
&= \underbrace{\text{Pr}(X = 0|T = 1)}_{=\frac{1}{4}} \cdot \underbrace{E[Y|T = 0, X = 0]}_{=800} + \underbrace{\text{Pr}(X = 1|T = 1)}_{=\frac{3}{4}} \cdot \underbrace{E[Y|T = 0, X = 1]}_{=1000} \\
&= 200 + 750
\end{aligned}$$

$$E[Y(0)|T = 1] = \boxed{\$950}$$

In summary:

- For $E[Y(0)]$: Average over $E[Y|T = 0, X = x]$ using $\text{Pr}(X = x)$.
- For $E[Y(0)|T = 1]$: Average over $E[Y|T = 0, X = x]$ using $\text{Pr}(X = x|T = 1)$.